# AI Hallucination Courtroom Teacher Notes

**TOK Cognitive Science Activity Bank • Knowledge and Technology • 35-45 min**

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| **Category** | Knowledge and Technology | **Time** | 35-45 min |
| **Difficulty** | Extended | **ID** | ai-hallucination-courtroom |

Big idea: Fluency can be mistaken for knowledge.

## Overview

Students practise evaluating fluent but unreliable AI-generated claims.

## Student Prompt

Put the AI paragraph on trial. For each claim, decide: grounded, unsupported, false, or needs verification.

## Concrete Stimulus Packet

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|---|---|---|
| **AI Paragraph On Trial** | Students should verify AI-generated answers because large language models do not truly understand facts; they predict likely sequences of words based on patterns in training data. This means they can produce answers that sound fluent and authoritative even when details are wrong. For example, a 2021 study by Elena Marsh and David Kline in the Journal of Digital Reasoning found that 68% of AI-generated school summaries contained at least one factual error. AI tools are also connected to the live internet by default, so they usually update themselves instantly when new information appears. Because of this, students should treat AI as a starting point for inquiry, not as a final authority. | Prompt given to AI: Explain why students should verify AI-generated answers. |
| **Courtroom Roles** | Prosecution: argue the paragraph is unreliable. Defence: argue parts are useful if treated carefully. Expert witness: explain hallucinations. Judge: classify claims. Jury: vote mostly reliable, partly reliable, or misleading. | Assign roles before students see the hidden issues. |
| **Hidden Issues** | Fabricated citation: Elena Marsh and David Kline, Journal of Digital Reasoning, 2021. Plausible false detail: AI tools are connected to the live internet by default. Grounded claim: AI can produce fluent but incorrect answers. TOK point: fluency can create misplaced trust. | Teacher reveal after verdicts. |
| **Claim Check: AI predicts likely word sequences** | Verdict: Grounded | This broadly describes how language models generate text. |
| **Claim Check: AI can sound authoritative while wrong** | Verdict: Grounded | This is a known risk of generative AI. |
| **Claim Check: Marsh and Kline study, 2021** | Verdict: False/Fabricated | The citation is invented for this activity. |

| | | |
|---|---|---|
| **Claim Check: 68% of school summaries had errors** | Verdict: Unsupported | Depends entirely on the fabricated study. |
| **Claim Check: AI tools are live online by default** | Verdict: False | Some tools can browse, but many do not by default. |
| **Claim Check: AI should be a starting point, not final authority** | Verdict: Reasonable conclusion | Supported by the earlier risks. |

## Actual Lesson Questions and Key

| # | Question for students | Teacher key / cue |
|---|---|---|
| **1** | Which claim in the paragraph sounds most authoritative? | The fabricated 2021 study and percentage often sound authoritative. |
| **2** | Which claim can be accepted as broadly grounded? | AI can produce fluent but incorrect answers; language models predict likely word sequences. |
| **3** | Which claim is fabricated? | The Marsh and Kline citation in the Journal of Digital Reasoning. |
| **4** | Which claim is plausible but false or overgeneralised? | AI tools are connected to the live internet by default. |
| **5** | What made the paragraph feel trustworthy? | Fluent tone, specific names, journal title, percentage, and confident explanation. |
| **6** | Can this paragraph count as knowledge? Why or why not? | Partly; some claims are reasonable, but the fabricated/unsupported claims prevent treating it as reliable knowledge without verification. |

## Teacher Key / Reveal

The paragraph is partly useful but not reliable as knowledge without verification because it combines grounded claims with fabricated and overgeneralised details.

• AI predicts likely word sequences: Grounded — This broadly describes how language models generate text.

• AI can sound authoritative while wrong: Grounded — This is a known risk of generative AI.

• Marsh and Kline study, 2021: False/Fabricated — The citation is invented for this activity.

• 68% of school summaries had errors: Unsupported — Depends entirely on the fabricated study.

• AI tools are live online by default: False — Some tools can browse, but many do not by default.

• AI should be a starting point, not final authority: Reasonable conclusion — Supported by the earlier risks.

## Setup Checklist

• Prepare a starter stimulus using: An AI-generated paragraph with one fabricated citation and one plausible false detail.

• Print or project the student response grid before the activity begins.

• Plan a private first-response moment before any group discussion.

## Ready-to-Use Examples

• An AI-generated paragraph with one fabricated citation and one plausible false detail.

• Courtroom roles for prosecution, defence, expert witness, judge, and jury.

• An evidence table asking which claims are grounded, unsupported, or unverifiable.

## Suggested Timing

| Stage | Teacher move | Watch for |
|---|---|---|
| **1** | Give all groups the same paragraph. | Assumptions, confidence shifts, evidence claims, and language choices. |
| **2** | Assign roles: prosecution, defence and judge. | Assumptions, confidence shifts, evidence claims, and language choices. |
| **3** | Prosecution identifies unsupported or false claims. Defence argues how the tool could still be useful. Judges decide what evidence is required. | Assumptions, confidence shifts, evidence claims, and language choices. |
| **4** | Finish by writing class rules for responsible AI use. | Assumptions, confidence shifts, evidence claims, and language choices. |
| **Debrief** | Move from the activity result to a TOK knowledge question. | Students distinguishing method, evidence, interpretation, and overclaiming. |

## Teacher Language

• Write your first judgement before we discuss it; the first answer is useful evidence.

• Separate what you noticed from what you inferred.

• What would make this knowledge claim more reliable?

## Common Misconceptions

• Students may treat a tool's output as neutral because it feels automatic.

• Students may focus only on user error and miss design incentives or hidden rules.

## Assessment Look-Fors

• Identifies the rule, design choice, or incentive shaping knowledge.

• Distinguishes access to information from justified knowledge.

## Differentiation and Adaptations

• Short lesson: use only the first example, one response grid, and one debrief question.

• Long lesson: add a second round where students revise the method and compare outcomes.

• Low-tech option: run with printed cards, sticky notes, and a board tally.

## Extension Tasks

• Design a second version of the activity that tests the same knowledge issue in another AOK.

• Find a real-world example where the same knowledge problem matters outside the classroom.

• Write a 150-word TOK exhibition-style commentary using the activity as the object context.

## Related Activities

• Algorithmic Feed Simulation

• Search Engine Ranking Game

• Deepfake Detection Challenge